

Differential Spectral Damping: Gap-Adaptive Regularization for Ill-Conditioned Kernel Methods

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Code: github.com/Praveg432/dsd-regularization

Problem

Least-Squares Twin SVM (LSTSVM) -- widely used for small-sample, high-dimensional classification in genomics, spectroscopy, and credit scoring -- requires inverting cross-class product matrices. At high dimensionality ($d \geq 100$), these matrices exhibit extreme spectral tail-clustering: the bottom 50% of eigenvalues compress into only 7% of the total spectral range. The Davis-Kahan theorem guarantees that eigenvectors corresponding to these clustered eigenvalues are directionally corrupted by numerical noise. Standard Tikhonov regularization applies a single uniform penalty that cannot resolve this structured corruption.

Proposed Method: Differential Spectral Damping (DSD)

DSD adapts regularization to the localized eigengap structure: $\lambda_i / (\lambda_i^2 + \alpha * \exp(-\beta * \delta_i))$, where δ_i is the minimum distance to adjacent eigenvalues. Large gaps (reliable eigenvectors) receive zero penalty; small gaps (corrupted) receive maximum suppression. Parameters α , β are initialized entirely from the spectral structure -- zero cross-validation needed.

Key Results (50-seed paired testing, fair baselines)

DSD (with NO optimization) vs Tikhonov (with 15-point grid search):

- GINA ($d=970$, real-world): +4.8pp, Cohen $d=4.49$, wins 30/30, $p < 0.0001$
- Madelon ($d=500$, real-world): +2.6pp, Cohen $d=1.76$, wins 27/30, $p < 0.0001$
- Synthetic $d=200$: +10.4pp, Cohen $d=1.57$, wins 44/50, $p < 0.0001$
- Synthetic $d=100$: +3.3pp, wins 35/50, $p = 0.0001$
- Synthetic $d=50$: No difference ($p=0.91$) -- operating boundary confirmed

Critical finding: Gradient-optimizing DSD degrades accuracy to Tikhonov-equivalent levels (85.9% $\rightarrow$ 81.1%), proving the advantage is a structural prior, not parametric.

Scientific Contributions

1. Novel eigengap-adaptive regularization formula motivated by Davis-Kahan theory
2. Empirical proof that structural spectral priors outperform parametric optimization
3. Clear operating regime characterization ($d \geq 100$, tail-clustering $< 12\%$)
4. Connection to algorithmic stability (soft spectral truncation $\rightarrow$ generalization)
5. Differentiable PyTorch implementation; computationally free ($O(m)$ overhead)

Significance

Cohen $d = 4.49$ on GINA is exceptionally large for a regularization technique ($d > 0.8$ is "large" by convention). DSD requires zero hyperparameter tuning while outperforming optimized baselines. Validated on two real-world OpenML benchmarks. Applicable wherever LSTSVM-type product matrices arise (genomics, spectroscopy, finance).

Limitations

- Does not help at $d \leq 50$ (insufficient tail-clustering)
- Does not beat Tikhonov on pre-image reconstruction (ties)
- No formal stability proof (conjectured, empirically validated)
- Not evaluated on non-RBF kernels

Target Venue

IEEE conference: ICASSP, ICPR, IJCNN, or IEEE Transactions on Neural Networks and Learning Systems (TNNLS).

Implementation

Complete open-source package: 10 source modules, 10 experiments, 28 unit tests. All results reproducible. Paper manuscript (6 pages, IEEE format) available.